

Part I - Identification

Identify the term described in each question. Answer keys appear beneath each question.

1. It is the way computers store, process, and express data using forms that the computer can understand.

Answer: Data Representation

2. These are numbers that can represent both positive and negative values.

Answer: Signed Numbers

3. It is the number system humans commonly use. It has 10 digits: 0-9. It is called base-10.

Answer: Decimal

4. It is a base-8 number system.

Answer: Octal

5. Method in converting binary to decimal where each position is worth double the value of the position to its right.

Answer: Doubling

6. The "brain" of the computer. The main processing component of a computer.

Answer: CPU

7. It is a communication pathway used to transfer information between components.

Answer: Bus

8. These are very small and extremely fast storage locations inside the CPU.

Answer: Registers

9. It describes how the internal parts of the CPU are arranged and how they work together to execute instructions.

Answer: CPU Organization

10. Common type of bus that carries data.

Answer: Data bus

Part II - True / False

Determine whether each statement is true or false. Answer keys appear beneath each question.

1. 1's Complement is a method of representing negative binary numbers by inverting every bit.

Answer: True

2. Program Counter (PC) stores the address of the next instruction.

Answer: True

3. In CPU organization, it means data automatically moves everywhere.

Answer: False

CPU organization does not mean that data automatically moves everywhere.

4. Each binary digit is called a byte.

Answer: False

Each binary digit is called a bit.

5. The ALU performs arithmetic and logical operations. It calculates numbers, performs logical operations, and manipulates binary data.

Answer: True

Part III - Conversion

Convert each number to the indicated number system. Answer keys appear beneath each question.

1. Convert 101_2 to decimal.

Answer: 5_{10}

2. Convert 10101_2 to decimal.

Answer: 21_{10}

3. Convert 1111101_2 to decimal.

Answer: 125_{10}

4. Convert 20_{10} to binary.

Answer: 10100_2

5. Convert 100_{10} to binary.

Answer: 1100100_2